

TALUKDAR RAIAN FERDOUS

SIGNAL PROCESSING & MACHINE LEARNING ENGINEER

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Education

University of Houston, USA

2021 - 2026

Doctors of Philosophy (Ph.D) in Biomedical Engineering.

- Masters completed; (CGPA: 3.668)

Khulna University of Engineering and Technology, Bangladesh

2015 - 2019

B.Sc. in Biomedical Engineering. (CGPA: 3.59)

Work Experience

University of Houston (Part-Time)

August 2021 – Present

Research Assistant

Houston, TX, USA

- Designed, developed and validated STM32 microcontroller based wearable neuroimaging device with biosensors: Electroencephalography (EEG), Functional Near-Infrared Spectroscopy (fNIRS), Accelerometer, Gyroscope for functional neuroimaging and multi-modal BCI applications. Implemented multi-system wireless data streaming using the Lab Streaming Layer (LSL) protocol.
- Formulated multi-modal time series analysis pipeline integrating EEG, fNIRS, heart rate and aortic blood pressure signals using MATLAB (EEGLAB, HOMER3, FieldTrip, NIRS-Toolbox) and Python (NumPy, Pandas, SciPy, Scikit-learn). Built EEG power-informed fNIRS general linear model (GLM) framework for passive BCI systems, enabling improved modeling of task-related neural activity. Built cross modal hybrid feature matrices and implemented machine learning models with rigorous cross-validation to evaluate performance. Communicated results through advanced data visualizations (Matplotlib, Seaborn, Plotly) highlighting measurable improvements from multimodal fusion.
- Implemented connectivity analysis for high density array iEEG data using Coherence, Phase Locking Value, Phase Lag Index, Pairwise Phase Consistency across multiple frequency bands (alpha, beta, theta, gamma) for auditory stimulus classification with deep learning algorithms (Neural Networks, Convolutional Neural Network, Long-short Term Memory) achieving 9.64% improvement over prior state-of-the-art studies.

Johnson & Johnson MedTech (Full-Time)

January 2025 – June 2025

Hemodynamics & Signal Processing Co-op

Danvers, MA, USA

- Applied data science and statistical analysis to large-scale clinical datasets from percutaneous ventricular assist devices (PVAD), processing Aortic Pressure (AoP) signal, Left Ventricular Pressure (LVP) signal and ECG signals stored on cloud-based internal databases. Utilized SQL queries through Python-based data pipelines to extract, filter, and preprocess device telemetry for downstream signal processing and modeling.
- Engineered large-scale clinical data processing pipelines and feature engineering algorithms for Impella Heart Pump logs, performed time and frequency based feature extraction and applied machine learning and deep learning algorithms using Pytorch and TensorFlow to improve the accuracy of real-time cardiac output prediction.
- Built and validated predictive data science models (Neural Networks, RNN) for continuous cardiac output (CO) estimation, incorporating physiological features such as respiratory-induced variability for enhanced interpretability and clinical decision support.
- Leveraged large language models (LLMs) by fine-tuning Llama 3.0 for automated adverse event text detection during clinical data collection, enabling scalable natural language processing (NLP) of unstructured physician notes and device logs to improve patient safety monitoring.

Bioforge Health Systems and Ltd (Full-Time)

January 2021 – July 2021

R&D Officer (Team Lead)

Dhaka, Bangladesh

- Researched and developed algorithms for wearable clinical vital signs monitoring system (ECG, SpO₂, heart rate, temperature) and formulated clinical data (MRI and Ultrasound) processing pipeline for a software assisting physician in decision making.
- Designed clinical study and executed equipment trials and quality assurance according to IEC (International Electrotechnical Commission) standards, ensuring the quality, safety, and compatibility of research equipment and procedures.
- Technical writing of manuscripts and research reports, communicating R&D outcomes to conferences, journals, and cross-functional stakeholders.

Projects

Brain-CAP High Density Neuroimaging System (NSF Funded) | *Python, Matlab, STM32 MC* August 2021 - Present

- Developed a groundbreaking high-density multimodal neuroimaging device in collaboration with National Science Foundation(NSF) and industrial partners (Brain Products and Indus Instruments) capable of collecting data from 64 EEG electrodes, over 1100 long-separation channels, and 256 short-separation fNIRS channels.
- Performed statistical test (paired t-test) using SciPy on resting-state data after pre-processing to validate brain signals comparing to a Gold standard fNIRS device. Conducted EEG frequency analysis to demonstrate strong alpha waves indicative of the resting state.

PHOEBE - fNIRS Data Quality Assessment Toolbox (Link) | *Matlab, Lab Streaming Layer* August 2021

- Co-developed a MATLAB-based toolbox for real-time functional near-infrared spectroscopy (fNIRS) data quality assesment, improving signal quality by extracting the cardiac band from raw signals and establishing quality metrics like scalp coupling index and peak spectral power.
- Automated the process of handling of real-time Lab Streaming Layer(LSL) streamed metadata from different manufacturer's devices by parsing the information with LSL library.

Dementia Risk Project (NIH Funded) | *Matlab, Gitlab Eprime* December 2023 - Present

- Testing and maintaining fNIRS device (NIRx NIRScout) and digitizer tool (Polhemus Patriot) utilized in data collection. Conducted routine inspections and troubleshooting, ensuring optimal performance and reliability of equipment throughout the research process.
- Conducted data quality assessment post data collection using QT-NIRS toolbox to increase accuracy in analysis and reliability of findings. Performed subject and group level statistical analysis (mixed effect model) to detect task sensitive brain regions in dementia riskd women with Type 2 Diabetes.

End of Life Project (MD Anderson Cancer Center) | *Matlab, EEGLAB, Clinical, R, Python* October 2022

- Pre-processed and analyzed clinical electro physiological data (EEG) using Matlab and Python to distinguish auditory responses in impending death patients. Conducted parametric and non-parametric statistical tests to identify statistically significant response against baseline noise at single subject level and visualized findings using color plots with the matplotlib library.
- Setup and test EEG equipment to record data from a near death population as well as developed an analytical framework capable of handling diverse data challenges such as poor quality, artifacts, and non-Gaussian distribution enabling robust data processing and interpretation.

Vital Signs Monitoring System Validation | *Matlab, Arduino, MongoDB* January 2021

- Developed algorithms for a wireless vital signs monitoring device capable of securely storing patient data, including heart rate, temperature, and ECG, to a secured server. Compared its performance to widely used medical devices for potential implementation in ICUs. Conducted data analysis using statistical tests to validate the device's reliability.

Publications

Journal Articles and Preprints

- [1] Electroencephalographic Signals of Auditory Perception in Unresponsive Patients in the Last Days of Life: Can You Still Hear Me? —**JAMA Neurology** 2026 (submitted)
- [2] EEG-Informed fNIRS Modeling Reveals Latent Multimodal Complementarity in an Aviation Cognitive Workload Dataset —**IEEE Transactions on Biomedical Engineering** 2026 (submitted)
- [3] Global Burden of 292 Causes of Death in 204 Countries and Territories and 660 Subnational Locations, 1990–2023: A Systematic Analysis for the Global Burden of Disease Study 2023—**The Lancet** 2025 (Link)
- [4] Global Age-Sex-Specific All-Cause Mortality and Life Expectancy Estimates for 204 Countries and Territories and 660 Subnational Locations, 1950–2023: A Demographic Analysis for the Global Burden of Disease Study 2023— **The Lancet** 2025 (Link)
- [5] Burden of 375 Diseases and Injuries, Risk-Attributable Burden of 88 Risk Factors, and Healthy Life Expectancy in 204 Countries and Territories, Including 660 Subnational Locations, 1990–2023: A Systematic Analysis for the Global Burden of Disease Study 2023— **The Lancet** 2025 (Link)
- [6] fNIRS reproducibility varies with data quality, analysis pipelines, and researcher experience— **Communications Biology Nature** 2025 (Link)
- [7] The fNIRS Glossary Project: A Consensus-based Resource for Functional Near-Infrared Spectroscopy Terminology — **Neurophotonics** 2025 (Link)
- [8] The fNIRS Reproducibility Study Hub (FRESH): Exploring Variability and Enhancing Transparency in fNIRS Neuroimaging Research— **MetaArXiv** 2025 (Link)

Conference Proceedings

- [1] Enhancing Auditory BCI Performance: Incorporation of Connectivity Analysis — **46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)** 2024 (Link)
- [2] Comprehensive Analysis on Feature Selection, Machine Learning and Deep Learning Algorithms to Detect Driver Drowsiness– An EEG Study — **Proceedings of the 10th International Conference on Biomedical and Bioinformatics Engineering by ACM (ICBBE)** 2023 (Link)
- [3] A Simple Design of a Matlab-Based Function for Topographical Presentation of fNIRS Data — **Proceedings of the 6th International Conference on Electrical, Control and Computer Engineering by Springer (ICECCE)** 2022 (Link)
- [4] A Cost-Effective Solution for Real-Time Remote Monitoring of Vital Signs in Patients — **14th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI) by IEEE** 2021 (Link)
- [5] Developing Cost Effective and Automated Saline Flow Rate Controlling System for Medical System — **International Conference on Electrical Information and Communication Technology (EICT)** 2019 (Link)
- [6] Biomaterials Serving the Purpose of Drug Delivery in Human Body — **5th International Conference on Mechanical, Industrial and Energy Engineering (ICMIEE)** 2018 (Link)
- [7] Available and Cost Reductive Materials in Bangladesh Having an Eye to Bone Treatment — **5th International Conference on Mechanical, Industrial and Energy Engineering (ICMIEE)** 2018 (Link)

Presentation and Workshop

Oral: Whole-head, Wearable, High-Density Integrated fNIRS-EEG Instrument for Naturalistic Studies (Link)

Poster: Standardization of fNIRS LSL Metadata Structure (SfNIRS 2022)(Link)

Workshop (Instructor): Software tools for assessing and enhancing quality of fNIRS measurements (SfNIRS 2022)
Introduction to fNIRS (IUCRC BRAIN Summer Program Workshop 2023)

Journal Reviewer: Neural Networks (Elsevier)- 21 Reviews,
Journal of Neuroscience Methods (Elsevier)- 1 Review,
Journal of Neural Engineering (IOP Science)- 4 Reviews,
Biomedical Physics & Engineering Express (IOP Science)- 1 Review

Certifications

Membership: IEEE Professional, Society for Neuroscience (sfN), The Society for Functional Near Infrared Spectroscopy(sFNIRS), Johnson Johnson Interns and Co-op Association(NA-ICA)

Programming: Introduction to Python Python Data Structure

Machine Learning: Machine Learning CNN with Tensorflow

Artificial Intelligence: AI in Healthcare Artificial Intelligence Symposium AI for Everyone

Clinical: Good Clinical Practice Biomedical Researcher Good Clinical Practice (US FDA Focus)

Bioinformatics: Bioinformatics